

Fifty Shades of Hue

Fully Automatic Color Mixer

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Introduction

This project aims to address the issue faced by beginner art students: the frequent waste of paint and inefficient color mixing process due to a lack of experience. We developed an automatic paint-mixing device: By integrating mechanical and automated technologies, including Turner poster colors, peristaltic pumps, a laser-cut frame, and a TCS34725 color sensor, the system allows users to simply scan a target color and receive a mixed watercolor paint.

Theoretical Basis: Kubelka-Munk

Direct linear mixing of RGB values often contradicts real-world color intuition, while Kubelka-Munk model considers the semi-empirical parameters to deal with this problem.

Given the absorption spectrum $K(\lambda)$ and scattering spectrum $S(\lambda)$ of each base pigment, and a mixing ratio vector $c = [c_1, c_2, \dots, c_N]$, the resulting mixed pigment's absorption and scattering properties can be modeled as weighted average of the individual pigments:

$$K_{mix}(c, \lambda) = \sum_{i=1}^N c_i K_i(\lambda), S_{mix}(c, \lambda) = \sum_{i=1}^N c_i S_i(\lambda)$$

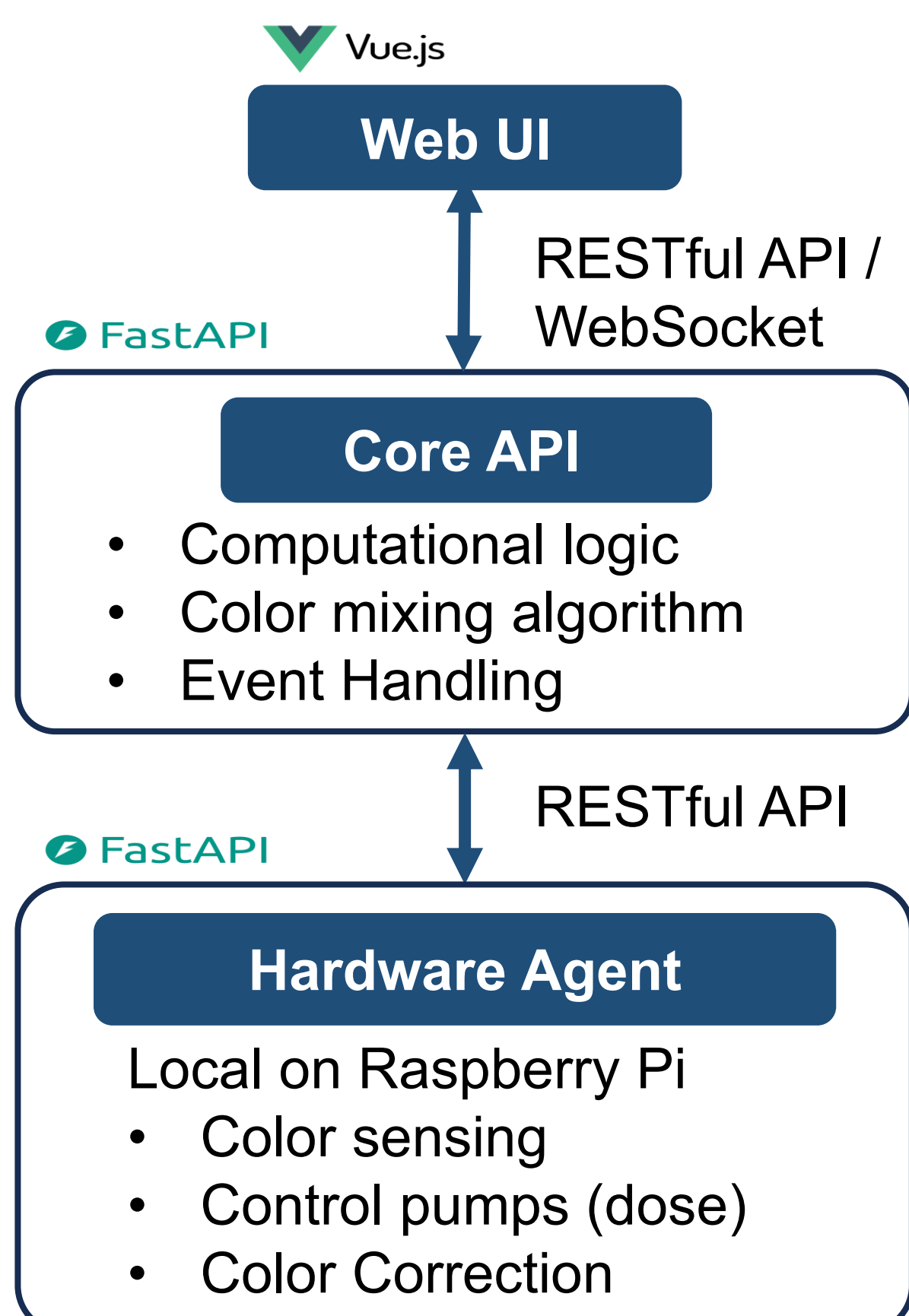
The predicted reflectance spectrum of the mixture is then given by

$$R_{mix}(c, \lambda) = 1 + \frac{K_{mix}(c, \lambda)}{S_{mix}(c, \lambda)} - \sqrt{\left(\frac{K_{mix}(c, \lambda)}{S_{mix}(c, \lambda)}\right)^2 + 2 \frac{K_{mix}(c, \lambda)}{S_{mix}(c, \lambda)}}$$

To further improve accuracy, a reflectance correction factor is introduced with two determined constants k_1, k_2 :

$$R'_{mix}(c, \lambda) = k_1 + \frac{(1 - k_1)(1 - k_2)R_{mix}(c, \lambda)}{1 - k_2 R_{mix}(c, \lambda)}$$

Software Structure



Hardware Components

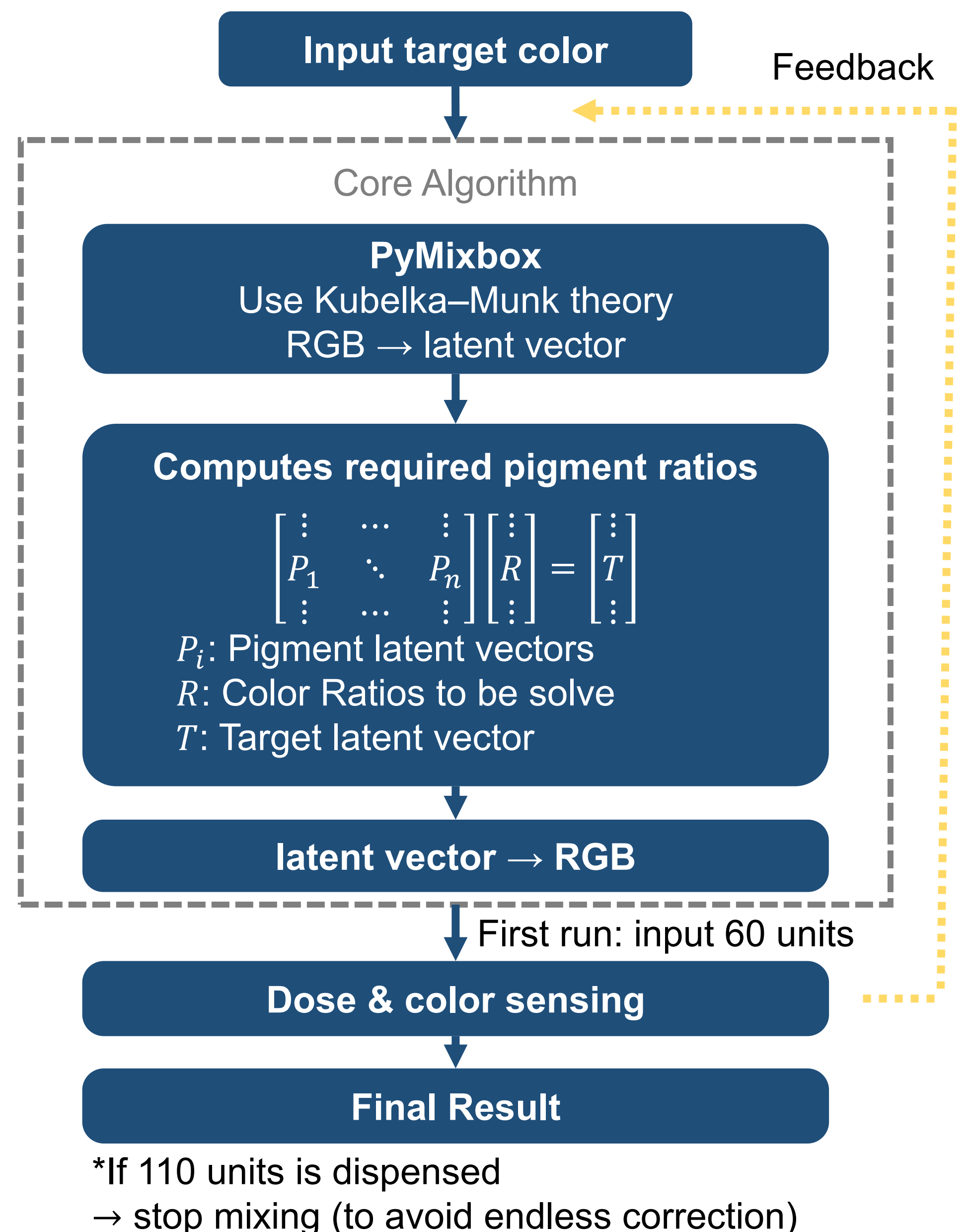
- Peristaltic pump
- Color sensor TCS34725
- Janua poster color + water
- Raspberry pi
- Power supply & relay



Color Sensor Calibration

- Fix integration time and light source before calibration.
- Dark subtraction and white balance.
- Color Correction Matrix (CCM): $VM = R \Rightarrow M = V^+R$
 V represents a color vector in the sensor's color space, whereas R denotes the real pigments' coordinates.
- Apply CCM to any new reading.
- Gamma correction: to display on screen.
Raise each channel (from 0 to 1) to the $\frac{1}{2.4}$ power, then multiply by 255 and round to get sRGB.

Process Flow



Prospects

- Color sensing improved for transparent pigments
→ mixing edible or natural dyes
→ food coloring or soap-making.
- Adapt the system for ceramic glaze mixing, where predicting color changes before and after firing remains a challenge.

